

Types and properties of Estimators

Wednesday, August 12, 2026 10:14 AM

- Estimators — Point Estimators
— Interval estimators

When is a point estimator good?

θ is a parameter which you estimate using $\hat{\theta}$

→ $\hat{\theta} \approx \theta$??

→ Can we decide using the data we collected??

① **Unbiasedness**: $\hat{\theta}$ is unbiased for θ if $E(\hat{\theta}) = \theta \quad \forall \theta \in \Omega_\theta$ **Definition**

Q. Is $\bar{X}$ an unbiased estimator of μ ?

$$E(\bar{X}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} n \mu = \mu$$

Q. Is s^2 an unbiased estimator of σ^2 ??

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2; \quad E[\chi_{n-1}^2] = n-1$$

$$E\left[\frac{(n-1)s^2}{\sigma^2}\right] = n-1 \Rightarrow \frac{n-1}{\sigma^2} E[s^2] = n-1 \Rightarrow E[s^2] = \sigma^2$$

Q. Is $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ an unbiased estimator for σ^2 ?

$$s^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$$

$$E[\hat{\sigma}^2] = E\left[\frac{n-1}{n} s^2\right] = \frac{n-1}{n} E[s^2] = \frac{n-1}{n} \sigma^2$$

Bias($\hat{\theta}$) = $E[\hat{\theta}] - \theta$ Definition

$$\text{Bias}(\hat{\sigma}^2) = \sigma^2 - \frac{1}{n} \sigma^2 - \sigma^2 = -\frac{1}{n} \sigma^2$$

Asymptotically unbiasedness is when $\text{Bias}(\hat{\theta}) \rightarrow 0$ as $n \rightarrow \infty$

$$E[\bar{X}] = \mu$$

$\bar{X}$ is unbiased for μ

$$E[\bar{X}] \neq \mu$$

$$E[\bar{X}] \neq \sigma^2$$

μ = avg. height
of campus
students

$n = 1000$

$$\begin{cases} \bar{X}_{\text{male}}, \bar{X}_{\text{male}} \rightarrow \bar{X} \approx \hat{\theta}_1 \\ \bar{X}_{\text{red}}, \bar{X}_{\text{red}} \rightarrow \bar{X} \approx \hat{\theta}_2 \end{cases}$$

$$\rightarrow \hat{\theta}_1^* \leftarrow \hat{\theta}_2$$

Standard error ($\hat{\theta}$) = s.d. ($\hat{\theta}$)

second property: Variance of a Point Estimator

We want to choose estimator that have smallest variance
but can we choose a constant?

Definition 9.2: If we consider all possible unbiased estimators of some parameter θ , the one with the smallest variance is called the **most efficient estimator** of θ .

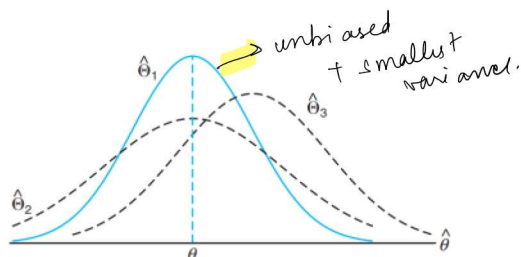


Figure 9.1: Sampling distributions of different estimators of θ .

generally, we pick estimator with minimum variance among the class of unbiased estimators

e.g. Normal population $X_i \sim N(\mu, \sigma^2)$

$$E[\bar{X}] = E[\tilde{X}] = \mu$$

?? mean median

$$\frac{\sigma^2}{n} = V(\bar{X}) < V(\tilde{X}) = \frac{\pi \sigma^2}{4n} ; \begin{matrix} n = \text{even} \\ n = 2m+1 \end{matrix} \rightarrow \begin{matrix} \frac{\pi}{4} \text{ even} \\ \frac{\pi}{4} \text{ odd} \end{matrix}$$

Mean Squared Error of $\hat{\theta}$ for θ

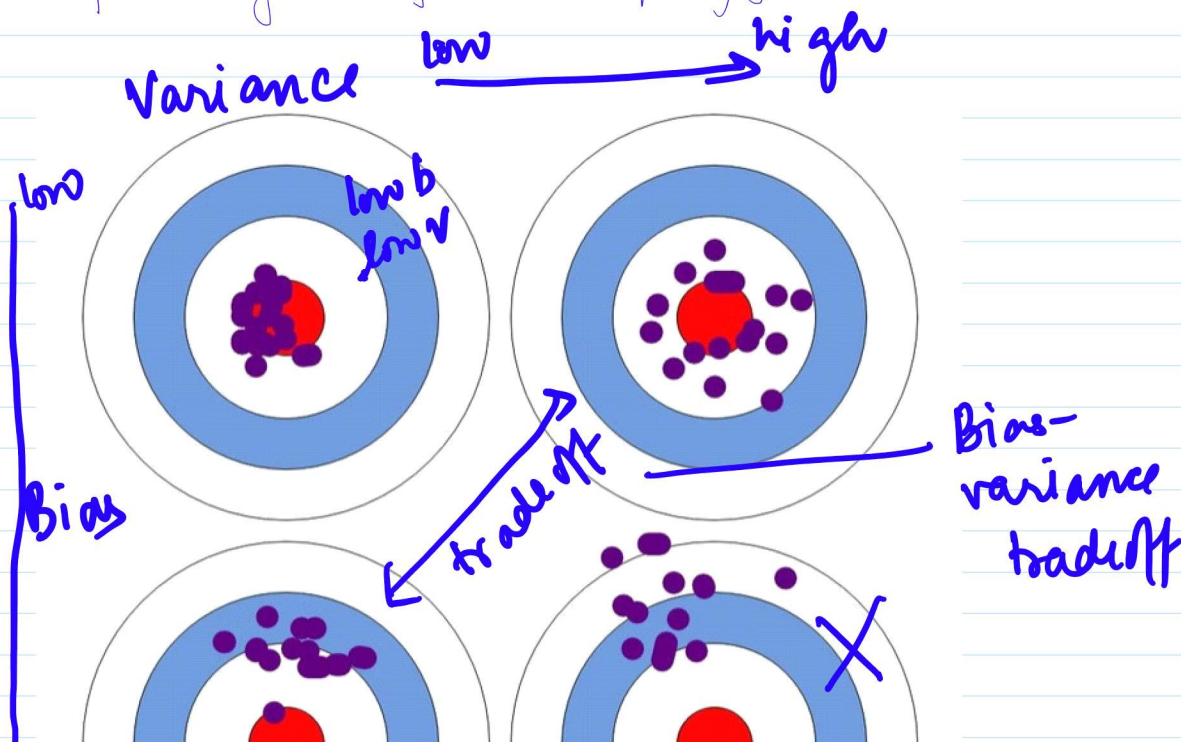
$$\begin{aligned} \text{MSE}(\hat{\theta}) &= E[\hat{\theta} - \theta]^2 = \text{Variance} + \text{Bias}^2 \\ &= E[\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta]^2 \\ &= E[\hat{\theta} - E(\hat{\theta})]^2 + E[E(\hat{\theta}) - \theta]^2 + 2E[(\hat{\theta} - E(\hat{\theta}))(E(\hat{\theta}) - \theta)] \\ &= \text{Var}(\hat{\theta}) + (E(\hat{\theta}) - \theta)^2 + 2E[\hat{\theta}E(\hat{\theta}) - (E(\hat{\theta}))^2 - \theta\hat{\theta} + \theta E(\hat{\theta})] \\ &= \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2 + 2[E(\hat{\theta})^2 - E(\hat{\theta})^2 - \theta E(\hat{\theta}) + \theta E(\hat{\theta})] \\ &\quad + 0 \end{aligned}$$

Consistency: An estimator T_n of parameter θ is **weakly** consistent if it converges in probability to θ , i.e.
 $\forall \epsilon > 0$

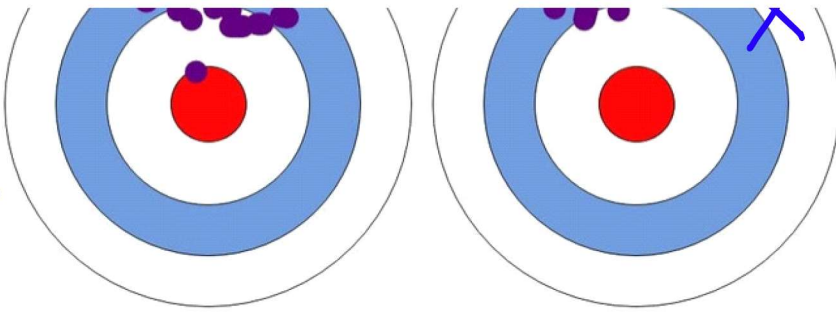
$$\lim_{n \rightarrow \infty} \Pr(|T_n - \theta| > \epsilon) = 0$$

e.g. $\bar{X}_n \xrightarrow{P} \mu$ [law of large numbers]

[proof using Chebyshev's inequality]



↓
high



Biased estimator but having small variance.
→ Lasso

Unbiased estimator + min variance
→ Least Squares (ideal)

Interval Estimator

Tuesday, September 9, 2025 11:46 AM

It is desirable to determine an interval within which we would expect to find the value of the parameter. Such an interval is called an interval estimate.

An interval estimate of a population parameter θ is an interval of the form $\hat{\theta}_L < \theta < \hat{\theta}_U$, where $\hat{\theta}_L$ and $\hat{\theta}_U$ depend on the value of the statistic $\hat{\Theta}$ for a particular sample and also on the sampling distribution of $\hat{\Theta}$. For example, a

avg. height of campus residents may have interval of the form $(4'5", 5'5")$

where $4'5''$ & $5'5''$ are computed using $\hat{\theta}$, n , s.e. $(\hat{\theta})$
 and estimators can be induced by their length. sampling $\text{dist}(\hat{\theta})$

Interval estimators can be judged by their lengths.

$\hat{\theta}_c, \hat{\theta}_u$ are dependent on θ

Since $\hat{\Theta}$ is a R.V., $\hat{\Theta}_1, \hat{\Theta}_n$ are also R.V.s.

We need to determine the values of $\hat{\theta}_1$ & $\hat{\theta}_n$

If $P(\hat{\Theta}_L < \theta < \hat{\Theta}_U) = 1 - \alpha$ for $0 < \alpha < 1$
then we have a probability of $1 - \alpha$ of selecting a random sample that will produce an interval containing θ .

$(\hat{\theta}_L, \hat{\theta}_U) \rightarrow (1-\alpha)\%$ confidence interval

$1 - \alpha \rightarrow$ confidence coefficient

$\hat{\theta}_L \rightarrow$ lower confidence limit

$\mathbb{Z}_n \rightarrow \mathbb{Z}_m$

$$\alpha = 0.05 \Rightarrow (1 - \alpha) \% = 95\%$$

$$\alpha = 0.01 \Rightarrow (1-\alpha)\% = 99\%$$

Which of 99% CI or 95% CI do you think is wider?

Smaller $\alpha \Rightarrow$ higher confidence $\Downarrow$

length of interval $\rightarrow$ larger

95% confident that avg. life of a product is between (6,7)
99% " " " " " " " " " " between (3,10)

95% confident that avg. life of a product is between (6, 7)
 99% " " " " " " " " between (3, 10)

Ideally, we prefer a short interval with a high degree of confidence.

$I_1 : (5'2", 5'4") \rightarrow 95\% \text{ CI}$
 $I_2 : (5', 5'6") \leftarrow 95\% \text{ CI}$

$\rightarrow$ length of the interval

$\rightarrow$ confidence coefficient α

95% CI		99% CI ✓	
(	)	(	)
(5'2", 5'4")	$\Leftarrow$ 95% CI		
(0.5', 10")	$\Leftarrow$ 100% CI		